

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
NASA SHARED SERVICES CENTER

RECOMMENDATION AND DETERMINATION TO SOLICIT FROM ONE SOURCE

I recommend that NASA, NASA Shared Services Center negotiate with Malvern Panalytical, Inc. only for service contract for the Empyrean x-ray diffraction system (SN 206771/DY1757). The total estimated cost of this effort is _____ and the period of performance is 3/15/2025 to 3/14/2026.

This recommendation is made pursuant to FAR 13.106-1(b)(1), for the acquisition of supplies or services determined to be reasonably available from only one source. Competition is impractical for the following reasons:

1. This service contract supports a NASA-owned Malvern Panalytical Empyrean X-ray Diffraction (XRD) system, a highly sensitive and complex instrument used to conduct crystallographic analysis of materials critical to aerospace research and development. Malvern Panalytical is the original equipment manufacturer (OEM) of the system, and due to the proprietary nature of the equipment and its components, specialized knowledge is required to maintain and service it properly.
2. Only Malvern Panalytical-trained service technicians possess the necessary expertise to service, align, and calibrate the system to ensure it delivers accurate data and maintains essential operational performance. Service documentation is proprietary, and all replacement parts are exclusively available through Malvern Panalytical. The service documentation is proprietary and replacement parts can only be provided by Tescan.

This instrument is a critical tool used for research programs for NASA. Improper maintenance and repair could cause extended down-time with a negative and costly impact to NASA programs.

Technical User Name

3/31/2025
Date

Title

Office